TECHNOLOGIES FOR MOTOR BEHAVIOUR ANALYSIS AND VIRTUAL MODELLING [1]

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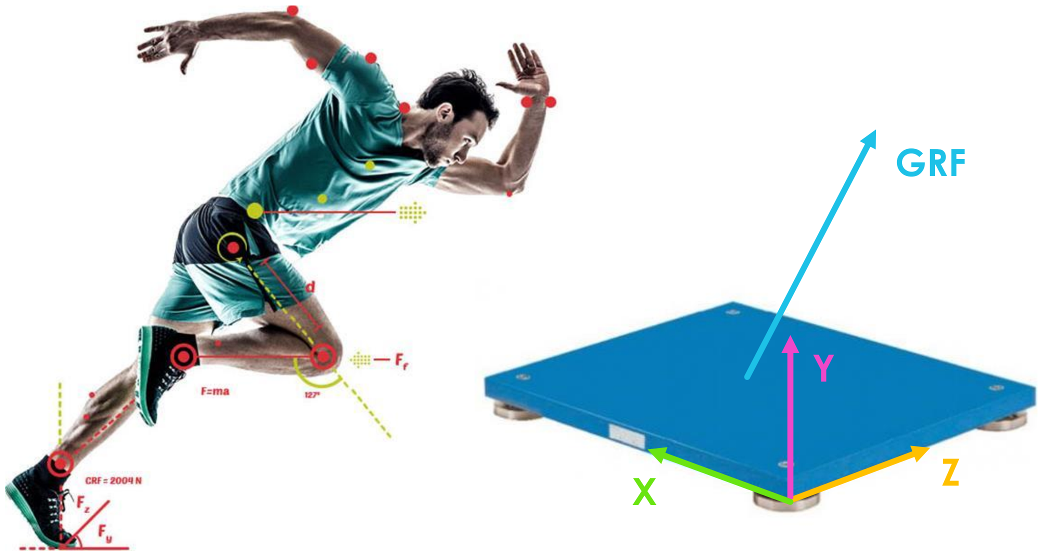
PRACTICE 3

Biomechanics and Force platforms

Laura Cruciani



LOAD CELLS

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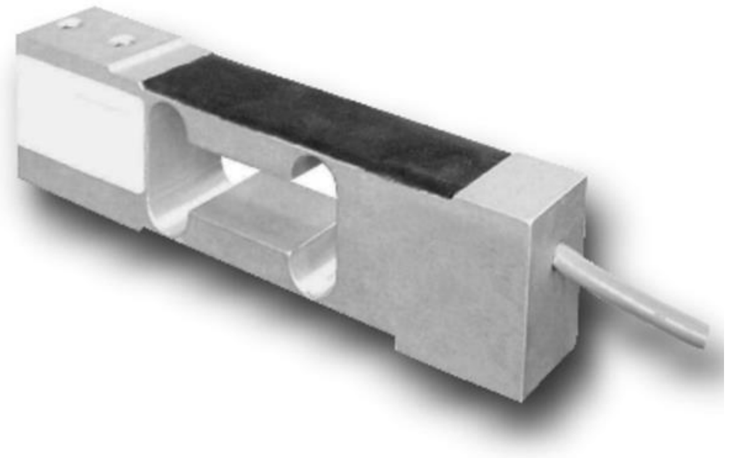
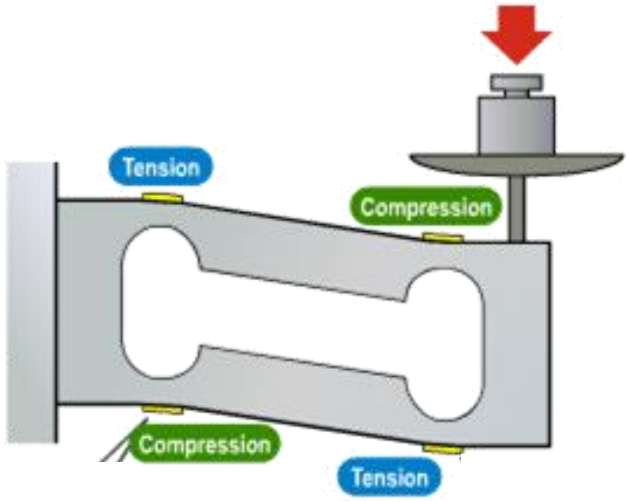
LOAD CELL

Force transducers that converts a **force** such as tension, compression, pressure or torque into **electrical signal**.

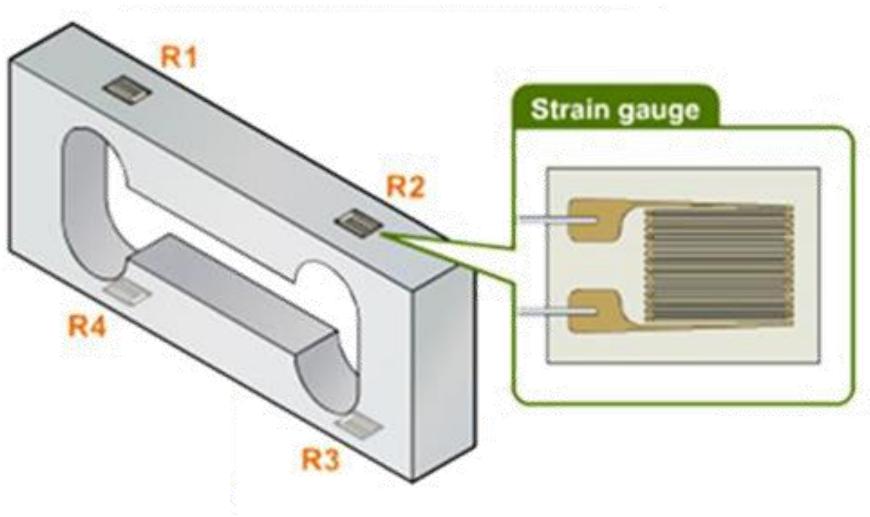


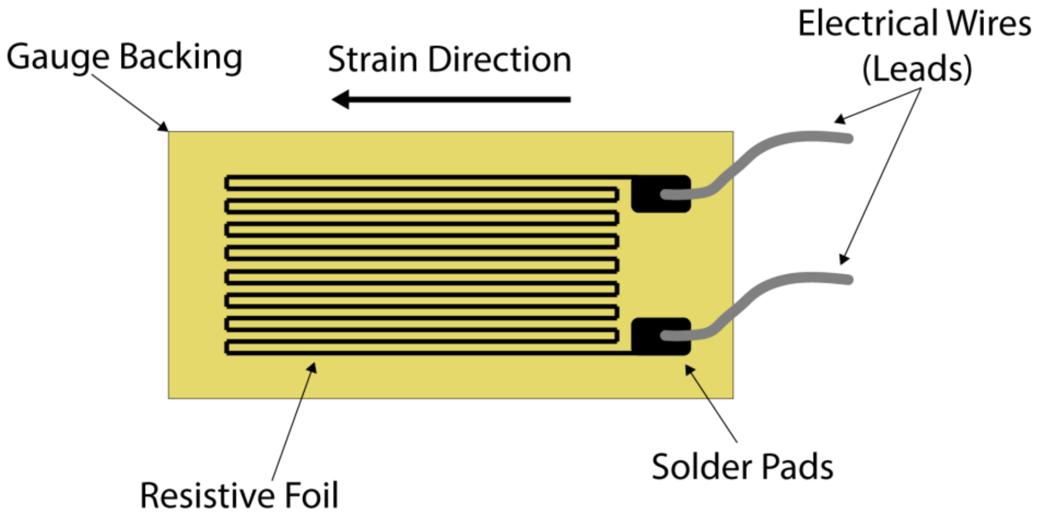


STRAIN GAUGE

STRAIN GAUGE

Device used to measure the **strain** on an object.
As the object deforms, the foil deforms causing its **electrical resistance** to change.





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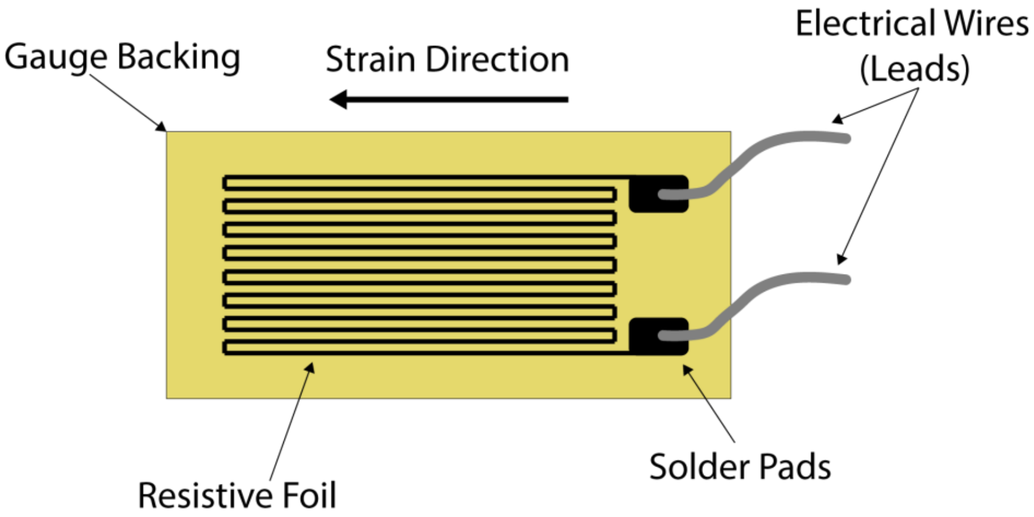

$$\boxed{\frac{dR}{R} = \frac{dL}{L} - \frac{dA}{A} + \frac{d\rho}{\rho}}$$

$$\boxed{G = \frac{dR}{R\varepsilon} \rightarrow \varepsilon = \frac{dl}{l}}$$

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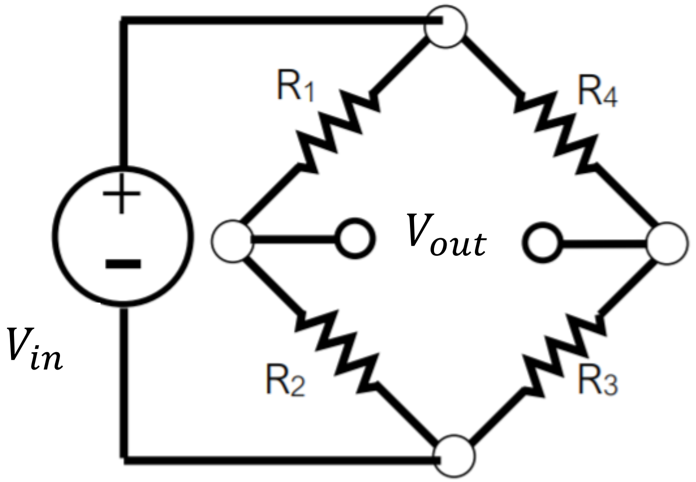
WHEATSTONE BRIDGE

The Wheatstone Bridge is an **electrical circuit** used to measure an overall **variation in resistance.**



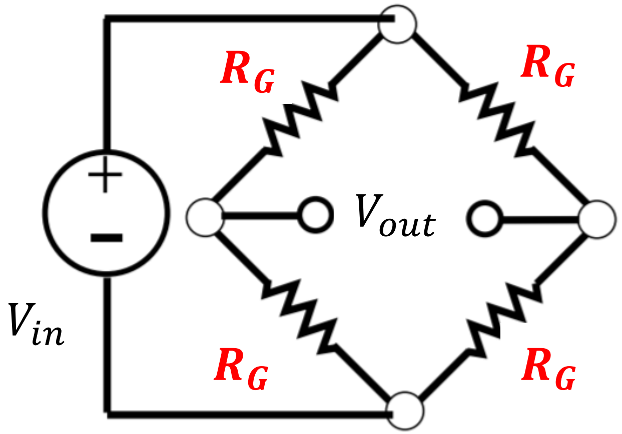
$$V_{out} = \frac{\Delta R}{R_0} * V_{in}$$

$$R_1 = R_0 + \Delta R$$
$$R_3 = R_0 + \Delta R$$
$$R_2 = R_0 - \Delta R$$
$$R_4 = R_0 - \Delta R$$

WHEATSTONE BRIDGE

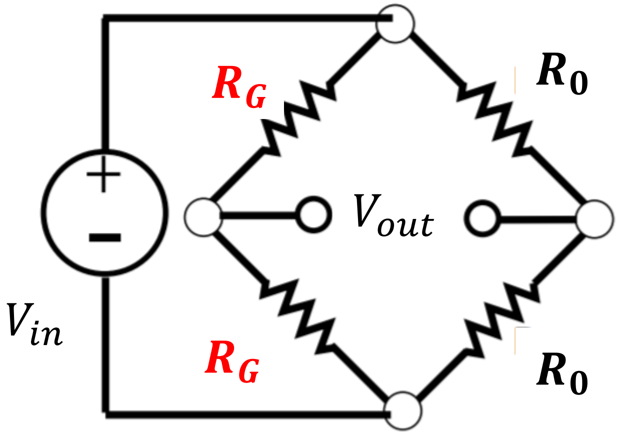
WHEATSTONE BRIDGE

Full bridge



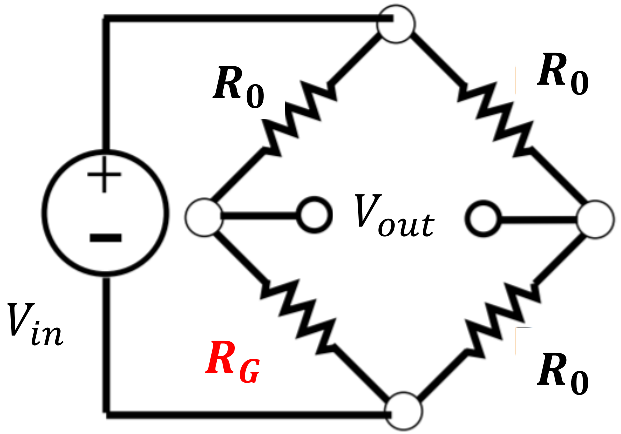
$$V_{out} = \frac{\Delta R}{R_0} * V_{in}$$

Half bridge



$$V_{out} = \frac{\Delta R}{2R_0} * V_{in}$$

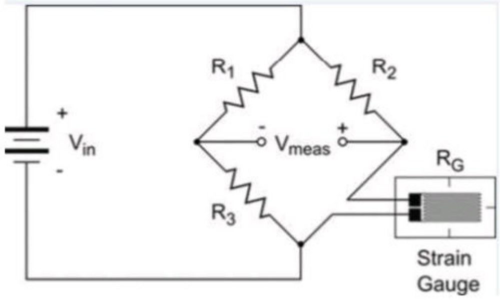
Quarter bridge



$$V_{out} = \frac{\Delta R}{4R_0} * V_{in}$$

EXERCISE 1

Strain Gauges are used to measure deformation by using resistance change.
Consider that Strain Gauge connected to **quarter-bridge** Wheatstone bridge.
If 5 Volts are applied as an input, what is the expected output voltage (V_{meas}) for 450 micro strain. Assume that Gage factor is 2.1 and R1=R2=R3=R0=120Ω.



- $V_{in} = 5V$
- $\varepsilon = 450\ \mu\ strain$
- $G = 2.1$
- $R_1 = R_2 = R_3 = R_0 = 120\Omega$

We have a quarter bridge. So, let's assume we're not applying any force so.

$$V^+ = V^- \qquad V_{meas} = 0\,V$$

During Compression $R_G < R_2 \rightarrow V^+ \downarrow$ cos $V^+ = V_{in} \cdot \frac{R_G}{R_G + R_2}$

so $V^+ > V^- \rightarrow V_{meas} < 0\,V$

During Tension $R_G > R_2 \rightarrow V^+ \uparrow$ Cos

So $V^+ > V^- \rightarrow V_{meas} > 0\,V$

Let's compute the V_{meas}

$$V_{meas} = V^+ - V^- \qquad V_{meas} = V_{in}\left(\frac{R_G}{R_2 + R_G} - \frac{R_1}{R_1 + R_3}\right)$$

if $R_1 = R_2 = R_3 = R_0$ and $R_G = R_0 + \Delta R$

$$V_{meas} = V_{in}\left(\frac{R_0 + \Delta R}{2R_0 + \Delta R} - \frac{1}{2}\right) = V_{in}\left(\frac{\Delta R}{4R_0 + \underbrace{2\Delta R}}\right)$$

$$V_{meas} \sim V_{in} \cdot \frac{\Delta R}{4R_0}$$ como $\Delta R \ll R_0$

since $G = \frac{\Delta R}{R_0 \cdot \varepsilon}$ $\Delta R = 2.1 \times 450 \cdot 10^{-6} \cdot 120 = 0.1134$

$$\boxed{V_{meas} = \frac{0.1134}{4 \times 120\Omega} \cdot 5V = 1.2\,mV}$$

PIEZOELECTRIC SENSORS

PIEZOELETRIC SENSORS

Versatile transducers that use the **piezoelectric effect** to measure changes in pressure, **force**, strain, acceleration and temperature.

PIEZOELECTRIC SENSORS

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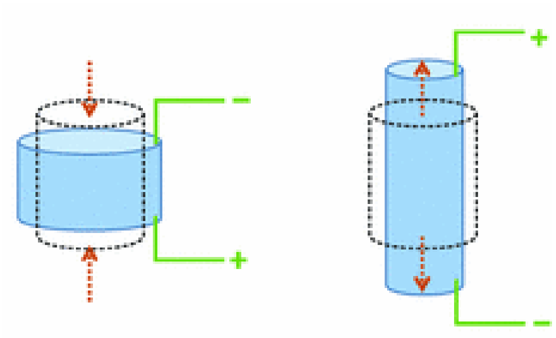
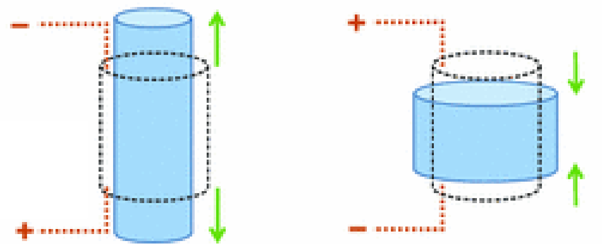
PIEZOELECTRIC EFFECT

Direct piezoelectric effect (sensor)
Change in electric polarization with a change in applied stress.



Input: strain
Output: voltage

Inverse piezoelectric effect (actuator)
Change of strain or stress in a material due to an applied electric field.



Input: voltage
Output: strain

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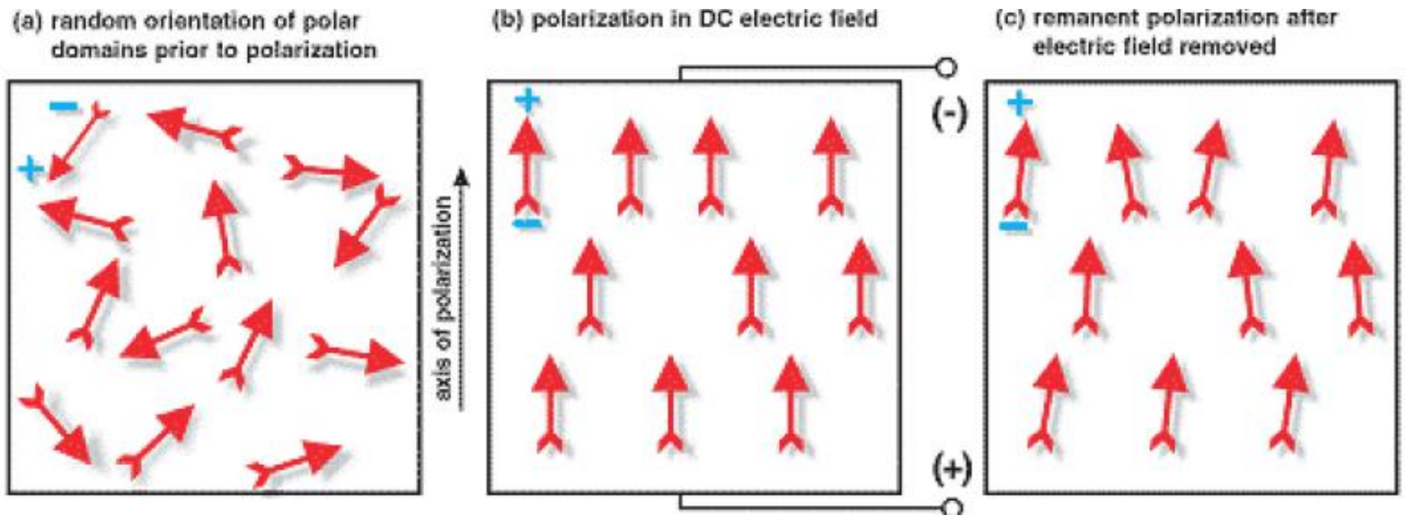
PIEZOELECTRICITY IN CERAMIC

The origin of piezoelectric effect is the displacement of ionic charges whitin a crystal structure (DIPOLE).

Poling (polarization): aligment of domains by exposing the element to a strong, static electric field at temperature below the *Curie point* (critical temperature).



configuration of near aligment.

Permanent polarization and permanently elongated

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PIEZOELECTRICITY IN CERAMIC

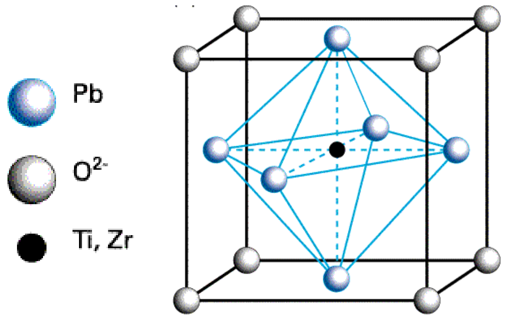
Poling (polarization): aligment of domains by exposing the element to a strong, static electric field at temperature below the *Curie point* (critical temperature).

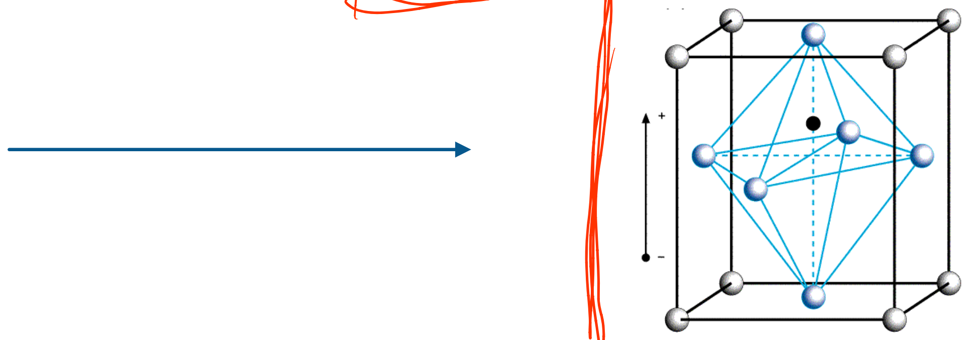
Above the *Curie point*:

- Symmetric charge distrubution
- Net electric pole moment is zero
- No piezoelectricity when deformed

Below the *Curie point*:

- Asymmetric charge distrubution
- Net polarization created
- Piezoelectricity when deformed





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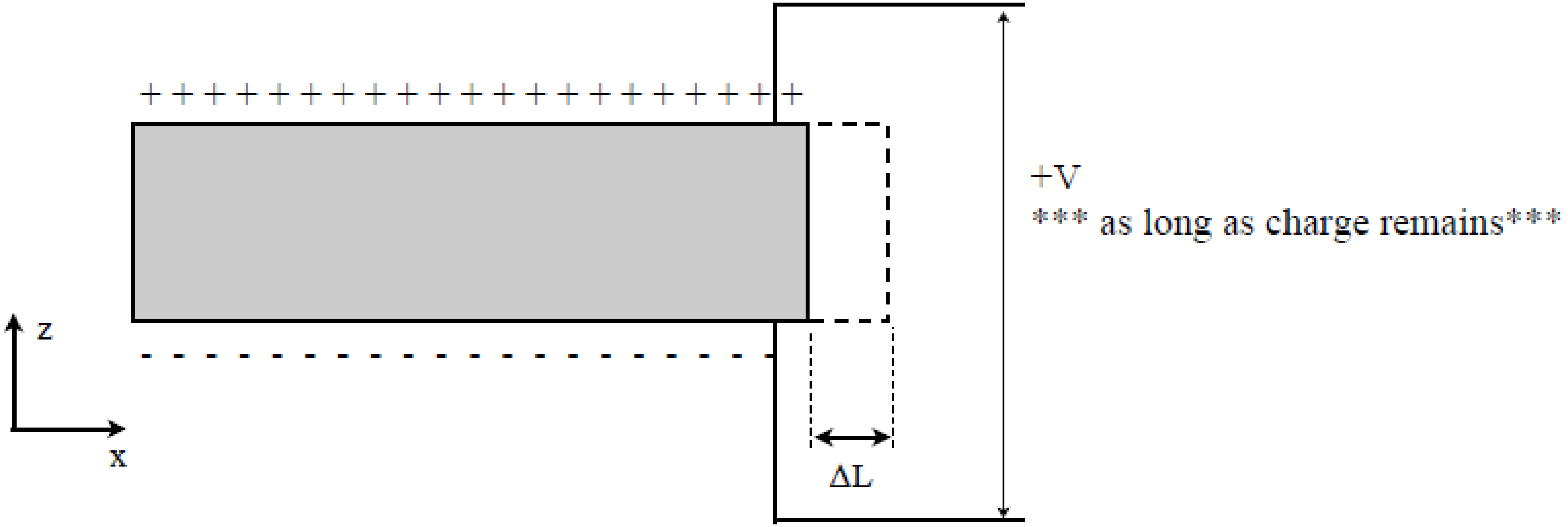
SIGNAL CONDITIONING

Piezoelectric crystall is like a capacitor that generates charge on the upper and lower surfaces when a strain is applied.

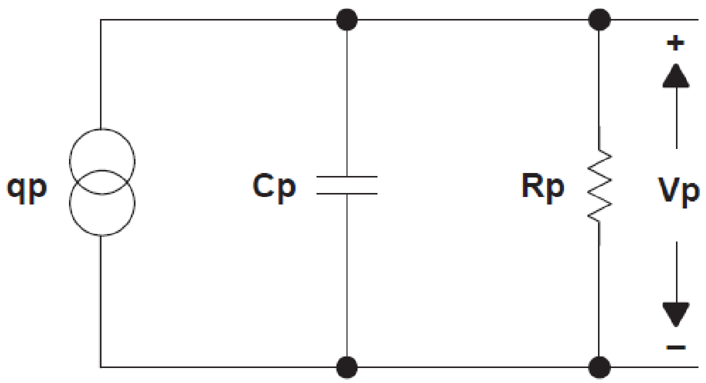


SIGNAL CONDITIONING

Capacitor that generates charge on the upper and lower surfaces when a strain is applied.

Charge source with shunt capacitor and resistor or voltage source in series with capacitor and resistor.

Charge Model



$$i = \frac{dQ}{dt} = k\frac{dp}{dt} = kP(s)s$$ → Current proportional to the derivative of pressure applied

$$G(s) := \frac{V_u(s)}{p(s)} = K\frac{sR_p}{1 + sC_pR_p}$$ → High pass at low frequences

K: device sensivity to pressure

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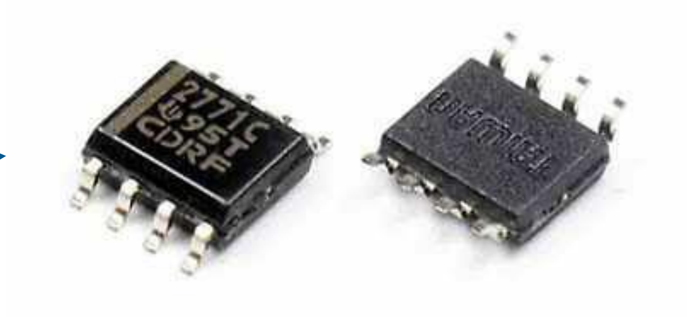
AMPLIFIER DESIGNING

- **Frequency** of operation
- **Signal amplitude:** desired signal levels between 3V-5V range full scale
- **Input impedance**: High input impedance
- Moderate **amount of amplification**
- **Mode of operation**:
  - Voltage mode
  - Charge mode → preferred



→ JFET or MOS input op amps are natural chioices (TLV2771)

cos of high input impedance.

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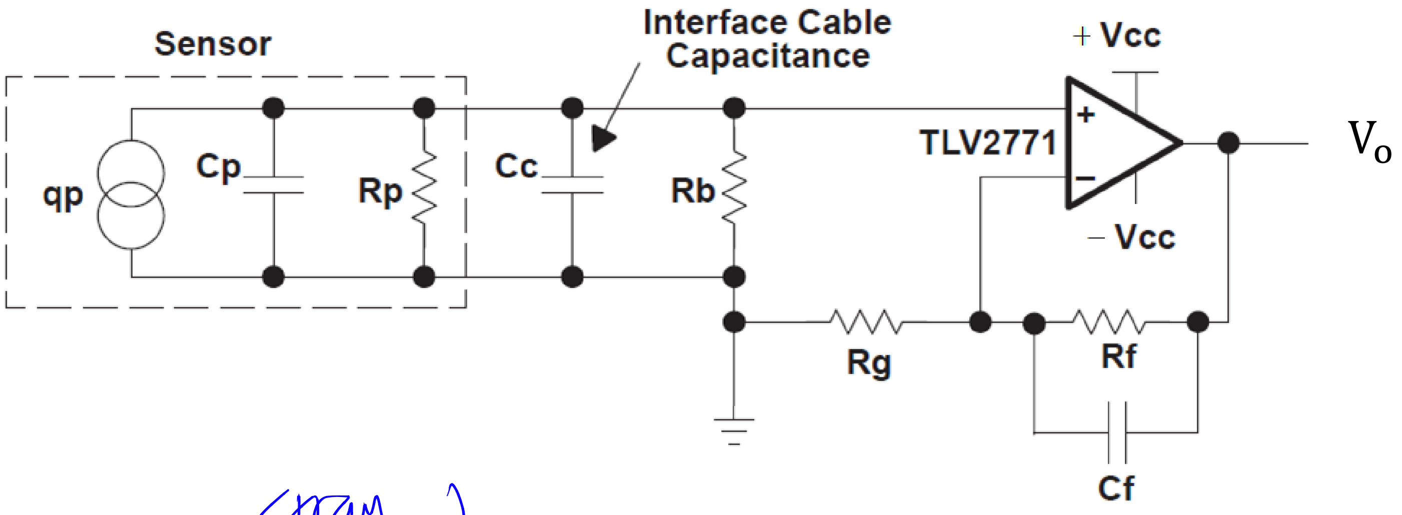
VOLTAGE MODE AMPLIFIER: used when the amplifier is very close to the sensor (small and stable cable capacitance).



$$V_o = \frac{qp}{(C_p + C_c)} \cdot \left[1 + \frac{R_f}{R_g}\right]$$

Output depends on the amount of capacitance seen by the sensor $(C_p + C_c)$

stray ?

Capacitance associated with the interface cable will affect the output voltage. If cable is moved or replaced, variations in Cc can cause errors.

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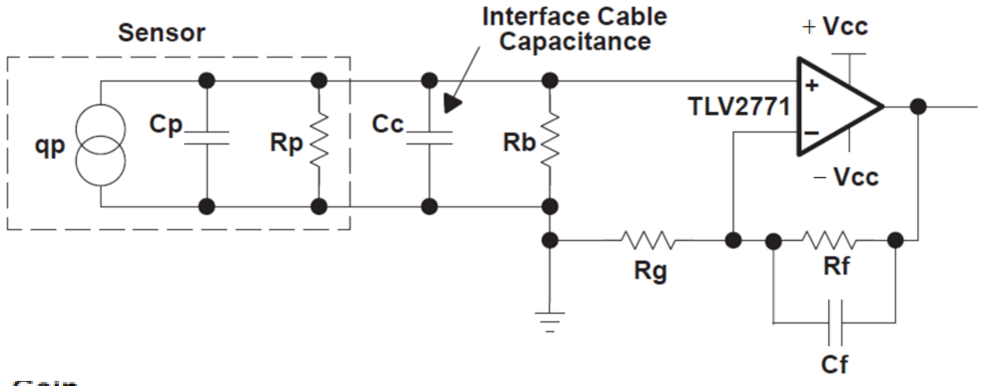
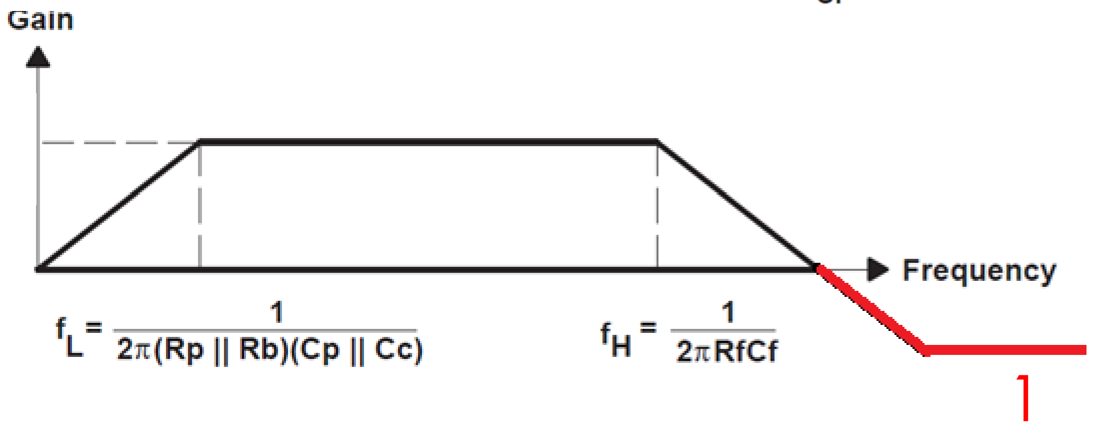
VOLTAGE MODE AMPLIFIER





$$V_o = \frac{qp}{(C_p + C_c)} \cdot \left[1 + \frac{R_f}{R_g}\right]$$

$$\text{Gain} = \frac{1}{(C_p + C_c)}\left(1 + \frac{R_f}{R_g}\right)$$

$$F_L = \frac{1}{2\pi(R_p||R_b)(C_p||C_c)} = \frac{R_p + R_b}{2\pi R_p R_b (C_p + C_c)}$$

$$F_H = \frac{1}{2\pi R_f C_f}$$

NOTE: the minimum gain is 1.

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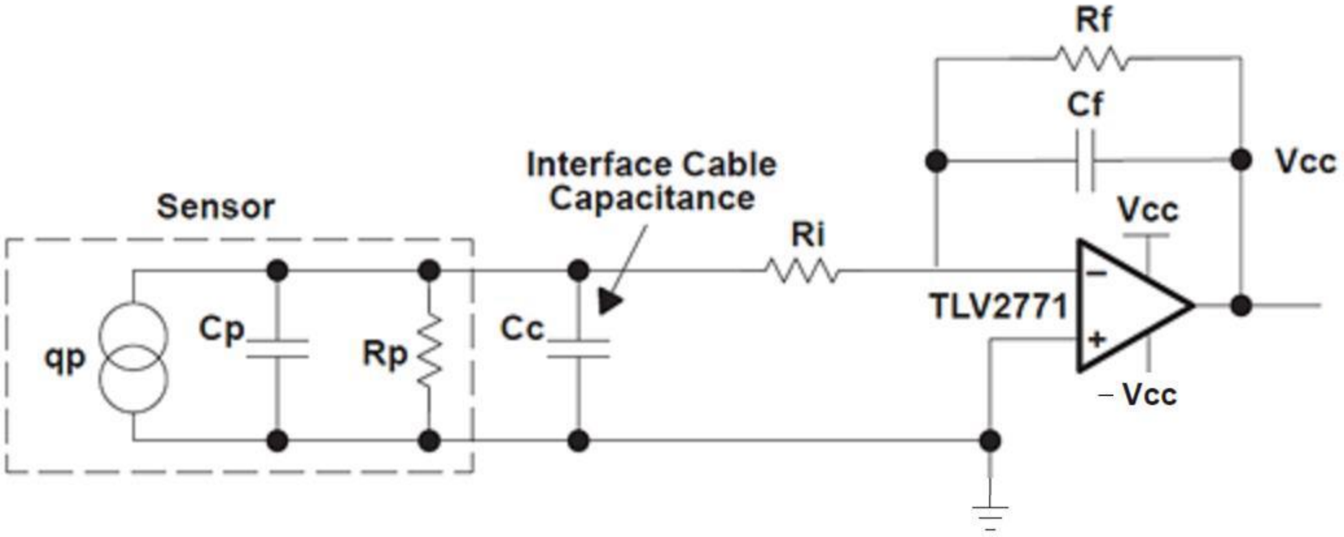
CHARGE MODE AMPLIFIER



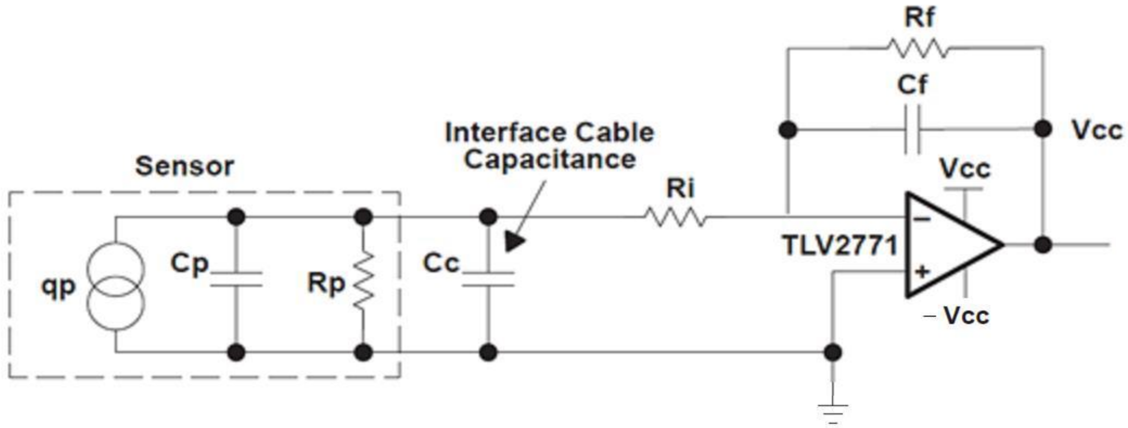
V_o

$$V_o = -\frac{qp}{C_f}$$

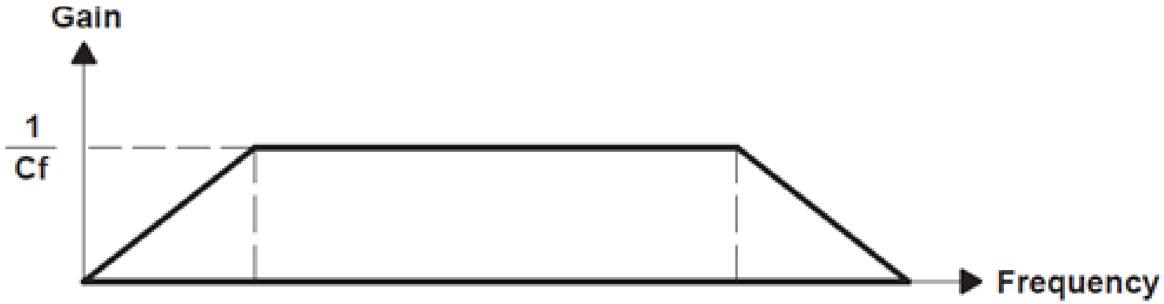
The amplifier mantains abut 0V across its input terminals. The stray capacitance associated with interface cabling does not present a big problem.

PIEZOELECTRIC SENSORS

CHARGE MODE AMPLIFIER



V_o



F_L F_H

$$\boxed{V_o = -\frac{qp}{C_f}}$$

$$\boxed{\text{Gain} = \frac{1}{C_f}}$$

$$\boxed{F_L = \frac{1}{2\pi R_f C_f}}$$

$$\boxed{F_H = \frac{R_p + R_b}{2\pi R_i (C_p + C_c)}}$$

Example: values

$F_L = 15.9\ \text{Hz}$ $C_f = 10^{-6}\text{F}$

$F_H = 1.59\ \text{kHz}$ $R_f = 10^{4}\Omega$

$C_p + C_c = 10^{-8}\text{F}$ $R_i = 10^{4}\Omega$

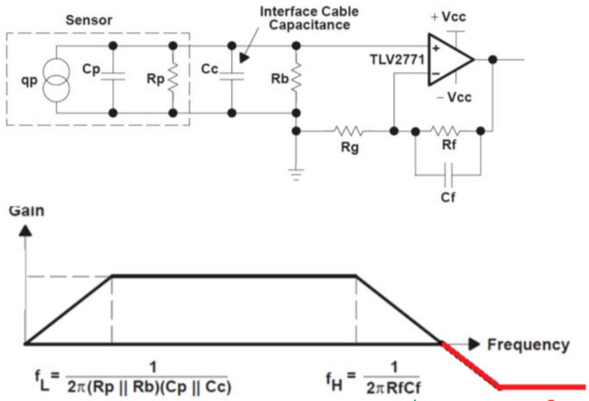
EXERCISE 2

You have been asked to design and develop a force platform conditioning circuit with a band-pass between 16Hz and 320Hz, knowing that:

- $k = 2.3\,\frac{pC}{N}$ (Force sensitivity)
- $C_p = 200pF$
- $R_p = 10^{10}\Omega$
- $R_i = 100\Omega$
- $R_b = 10M\Omega$

2.1. Use a voltage mode amplifier with a gain in the band-pass equal to 1.15 $\frac{V}{N}$ and $R_f = 5M\Omega$.

2.2. Use a charge mode amplifier with a gain in the band-pass equal to 0.0023 $\frac{V}{N}$.



$$F_L = \frac{1}{2\pi(R_p \| R_b)(C_p \| C_c)}$$

16 Hz

10^{10} 10MΩ 200pF ?

$$F_L = \frac{R_p + R_b}{2\pi\, R_p \cdot R_b \cdot (C_p + C_c)}$$

$$C_c = \frac{R_p + R_b}{2\pi R_p R_b F_L} - C_p = \boxed{795\ pF}$$

$$F_H = \frac{1}{2\pi R_f C_f}$$

320 Hz 5MΩ ?

$$C_f = \frac{1}{2\pi R_f F_H} = \boxed{99.5\ pF}$$

$$\boxed{\text{Gain} = \frac{K}{(C_p + C_c)}\left(1 + \frac{R_f}{R_g}\right)}$$

2.3 pC/N

5 MΩ

1.15 200 pF 795 pF ?

$$\boxed{R_g = 10K\Omega}$$

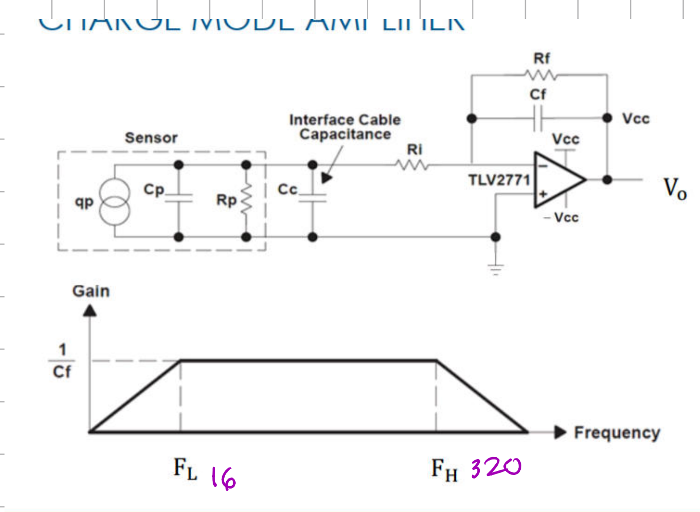
Now, using charge amplifier with a gain of 0.0023 V/N

$$V_o = -\frac{qp}{C_f}$$

$$\text{Gain} = \frac{K}{C_f}$$

$$F_L = \frac{1}{2\pi R_f C_f}$$

$$F_H = \frac{1}{2\pi R_i (C_p + C_c)}$$



$$F_H = \frac{1}{2\pi R_i (C_p + C_c)}$$

F_H: 320

R_i: 100

C_p: 200pF

C_c: ?

$$\boxed{C_c = 5\mu F}$$

$$\text{Gain} = \frac{K}{C_f}$$

Gain: 0.0023 V/N

K: 2.3×10^{-12}

C_f: ?

$$\boxed{C_f = 1nF}$$

$$F_L = \frac{1}{2\pi R_f C_f}$$

F_L: 16

R_f: ?

C_f: 1nF

$$\boxed{R_f = 9.95\,M\Omega}$$

FORCE PLATFORMS

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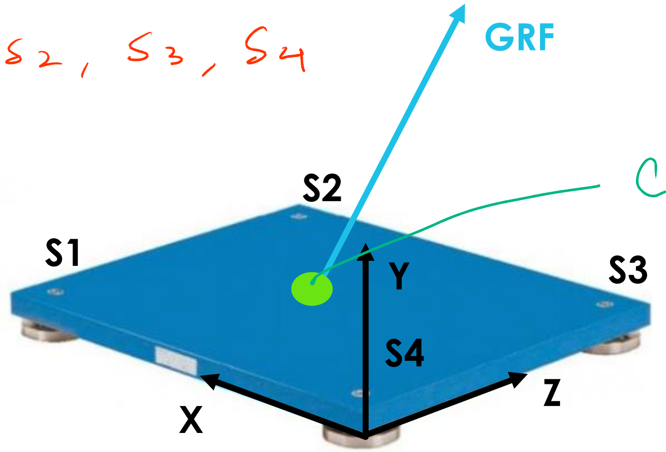
FORCE PLATFORMS: metal platforms in which force transducers are embedded used to measure force in 3D (**Ground Reaction Forces – GRFs**).

Especially

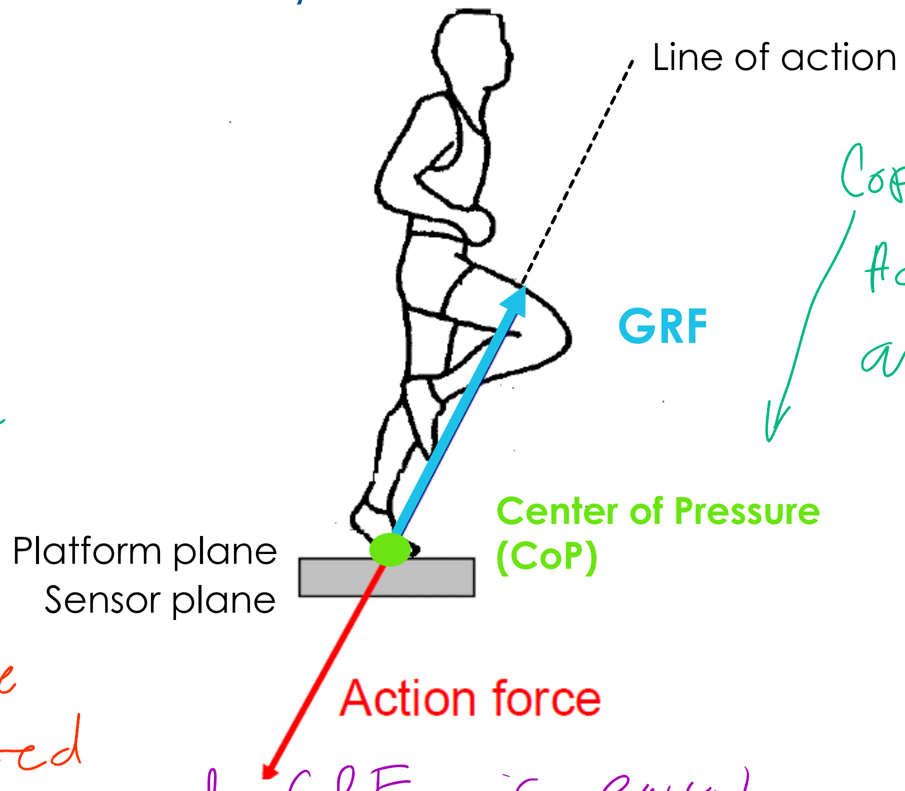
Four sensors s_1, s_2, s_3, s_4



Center of Pressure

At least 3 sensors

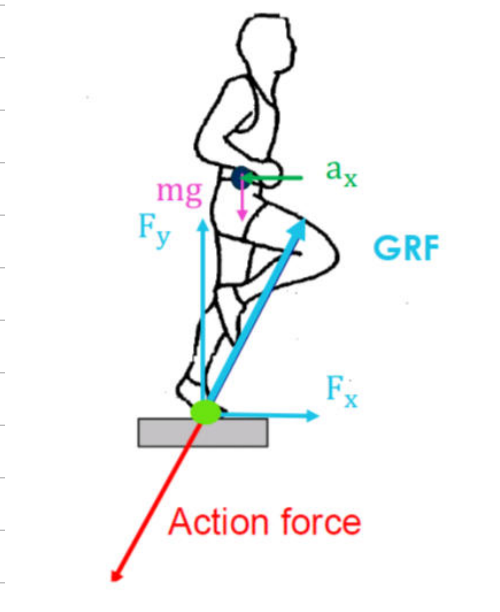
Action Force is generated when step



CoP is where Action force and platform meet

and GRF is equal

Here we can find the ground reaction forces by discomposing this force



A 2D approximation is done

The subject is walking/running, thus an inertial acceleration along x and we need to consider the weight of the subject

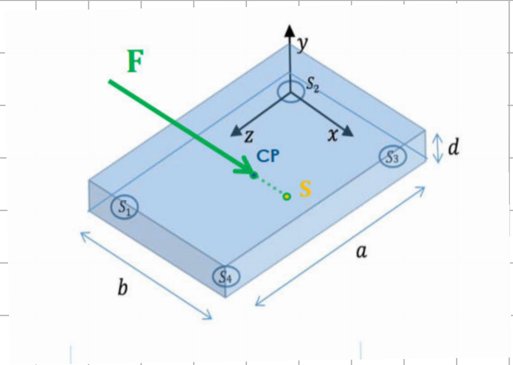
$$\sum F_x = m \cdot a_x \rightarrow F_x = m \cdot a_x$$

$$\sum F_y = m \cdot a_y \rightarrow F_y - m \cdot g = 0$$

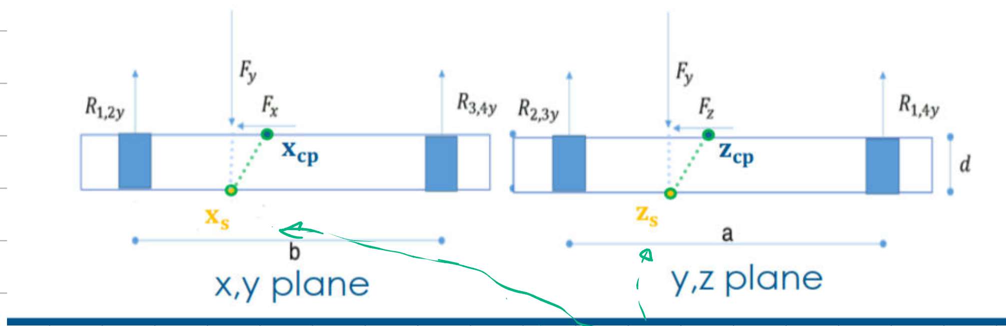
$$\hookrightarrow F_y = m \cdot g$$

no acceleration in y but weight is considered

To find the center of pressure



We need to work this 3D problem into an 2D problem



we need to compute the point S → intersection between ground reaction force F and sensor plane (lower surface)

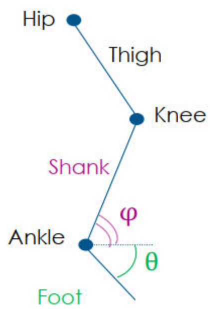
2. Compute the point (S) of intersection between F and Sensor plane (lower surface)
   - Equilibrium of forces $\sum F = m \cdot a$ → inertia
   - Equilibrium of moments $\sum M = I \cdot \alpha$ → angular acceleration

Once we know x_s and z_s we can find C_p

3. Compute the point (CP) of intersection between F and the platform surface

EXERCISE 3

Consider a subject weighting 80Kg, for whom the distance between ankle and foot is 20cm and the length of the shank is 45cm. The angle θ between the foot and the horizontal axis is equal to 80° and the one between the shank and the horizontal axis (φ) is 42°.



$m = 80Kg$
$l_{foot} = 20cm$
$l_{shank} = 45cm$
$\theta = 80°$
$\varphi = 42°$

Segment	$\frac{Mass_{segment}}{Mass_{total}}$	$\frac{CoM}{l}$ Prox	Dist	$\frac{Gyradius}{l}$
Foot	0.0145	0.5	0.5	0.47
Shank	0.064	0.433	0.567	0.302
Thigh	0.1	0.433	0.567	0.323

Ratio between center of mass/Length

Foot, Shank and thigh are connected between them with these three joints: Hip, Knee, Ankle

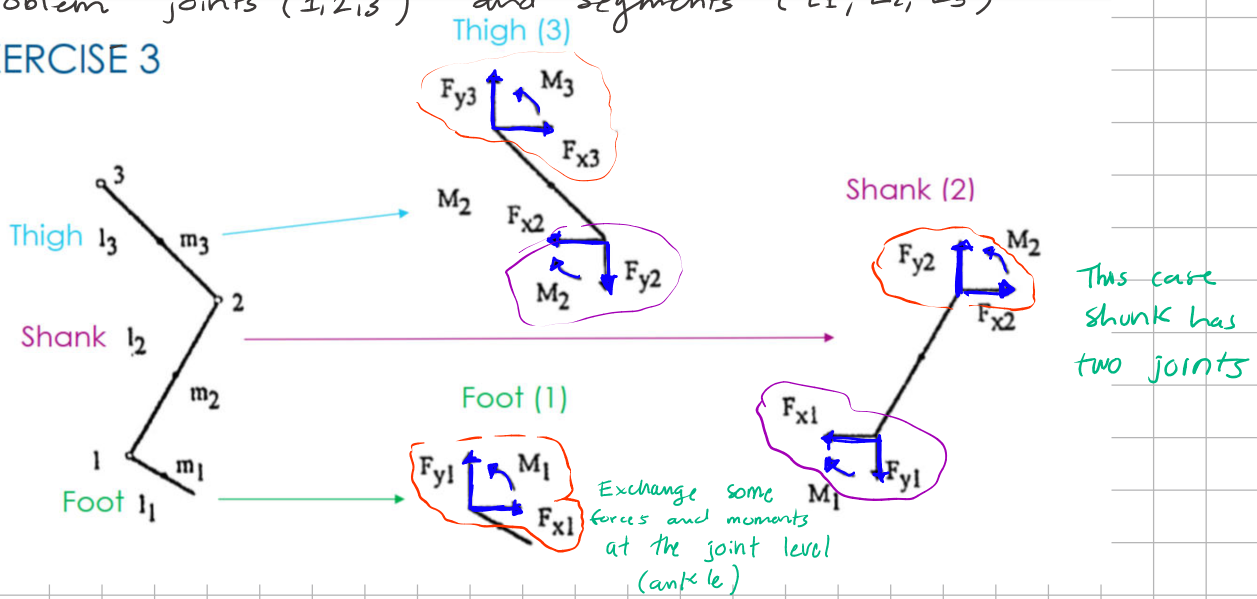
The center of mass is the point where the mass is concentrated

Gyradius is the distance between the axis of rotation of a segment and the point where all the mass is concentrated

This is needed to compute the inertia

We start by decomposing the problem joints (1,2,3) and segments (L_1, L_2, L_3)

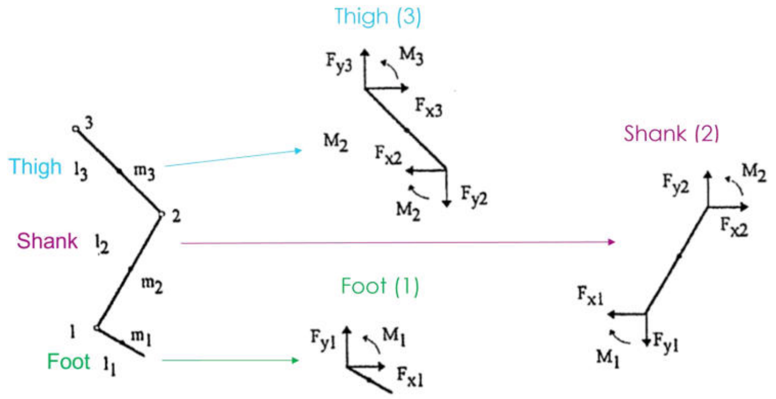
EXERCISE 3



REQUEST AND SOLUTION

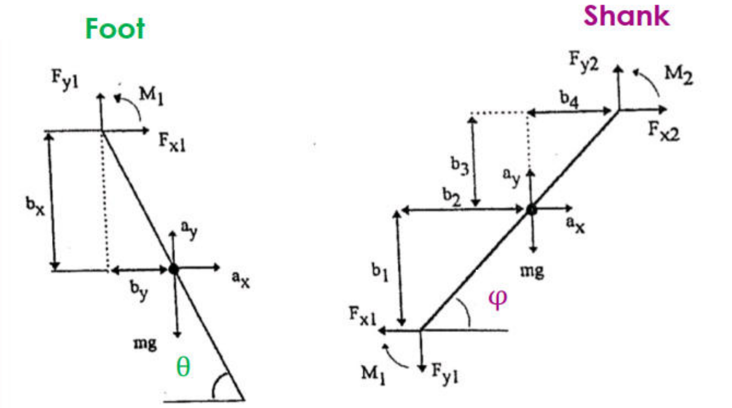
3.1. Compute the muscular moment and the reaction forces at the ankle and knee, during the oscillaltion of foot and tibia, when the accelerations of the COMs are:

	Foot	Shank
$a_x \left[\frac{m}{s^2}\right]$	9	-0.2
$a_y \left[\frac{m}{s^2}\right]$	-6.65	-4.2
$\alpha \left[\frac{rad}{s^2}\right]$	22	35



So, only foot and shank

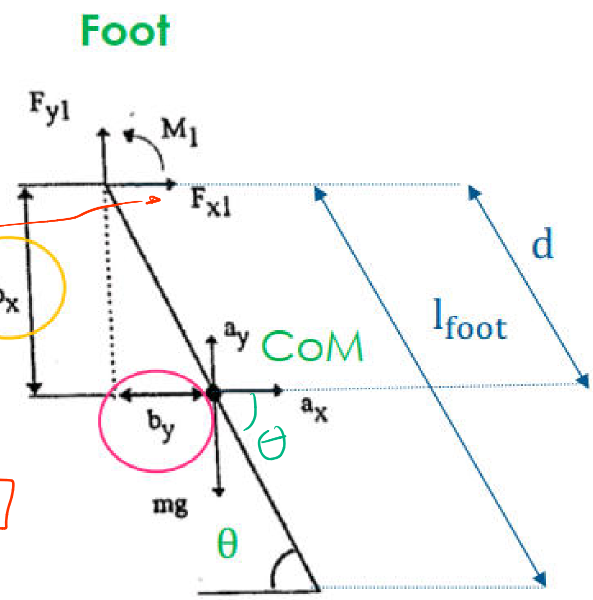
	Foot	Shank
$a_x \left[\frac{m}{s^2}\right]$	9	-0.2
$a_y \left[\frac{m}{s^2}\right]$	-6.65	-4.2
$\alpha \left[\frac{rad}{s^2}\right]$	22	35



→ mass segment / total mass

$m_{foot} = 0.0145 \cdot m = 0.0145 \cdot 80$

$m_{foot} = 1.16\,Kg$



Then we need to solve equilibrium of forces and moments w.r.t x and y components

F_{x1}

$\sum F_x = m_{foot}\, a_x \rightarrow 1.16\,Kg * 9 = \boxed{10.44N}$

$\sum F_y = m_{foot}\, a_y$

↳ $F_{y1} - m_{foot} \cdot g = m_{foot} \cdot a_y$

$F_{y1} = m_{foot}(g + a_y) = \boxed{3.634\ N}$

$\sum M = I_{foot} \cdot \alpha$

we consider CoM as reference point and +↺

Momento producido por la fuerza vertical en CoM

$M_1 = F_x b_x + F_y b_y + I_{foot} \cdot \alpha$

momento producido por la fuerza horizontal en CoM

REQUEST AND SOLUTION

proxi distal

$d = 0.5 \cdot l_{foot} = 10cm$

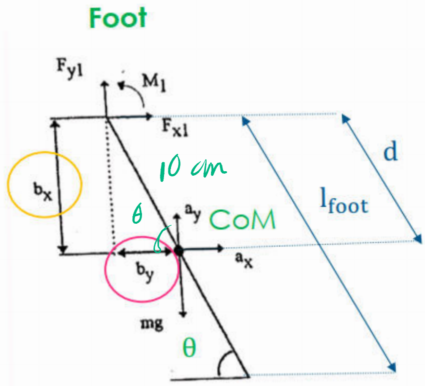
$b_x = d \cdot \sin(\theta) = 9.8cm$

$b_y = d \cdot \cos(\theta) = 1.7cm$

Inertia: $I_{foot} = m_{foot} \cdot r^2$ ← Gyradius gives the relation between r / L.

$CoG = \frac{Gyradius}{l} = 0.47 \rightarrow r = CoG \cdot l_{foot} = 0.47 \cdot 0.2 = \boxed{9.4cm}$

$I_{foot} = 1.16kg \cdot (9.4 \cdot 10^{-2})^2 = 0.0103kg \cdot m^2$



Coming back we have

$$M_1 = \underbrace{Fx}_{10.44}\, \underbrace{bx}_{9.8\,cm} + \underbrace{Fy}_{3.654}\, \underbrace{by}_{1.7cm} + \underbrace{I_{foot}}_{0.0103} \cdot \underbrace{\alpha}_{22} = \boxed{1.312 \; N \cdot m}$$

$M_1 > 0$ Ankle dorsiflexion thanks to tibialis anterior

Now the same with the shank

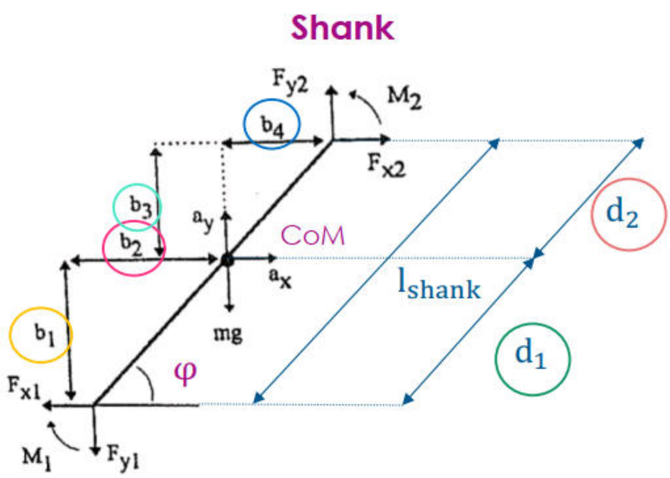
Segment	$\frac{Mass_{segment}}{Mass_{total}}$	$\frac{CoM}{l}$ Prox	Dist	$\frac{Gyradius}{l}$
Foot	0.0145	0.5	0.5	0.47
Shank	0.064	0.433	0.567	0.302

$d_1 = 0.567 * l_{shank} = 25.515 cm$

(distal - ankle - com)

$d_2 = 0.433 * l_{shank} = 19.485 cm$

proximal CoM - Knee



And with d_1 and d_2 we get.

$b_1 = d_1 \cdot \sin(\varphi) = 77.1 \text{ cm}$ $\quad$ $b_3 = d_2 \cdot \sin(\varphi) = 13 \text{ cm}$

$b_2 = d_2 \cdot \cos(\varphi) = 18.96 \text{ cm}$ $\quad$ $b_4 = d_2 \cdot \cos(\varphi) = 14.5 \text{ cm}$

Then we have the mass

$$m_{shank} = 0.0465 \cdot m = 0.0465 \cdot 80 = 3.72 \text{ Kg}$$

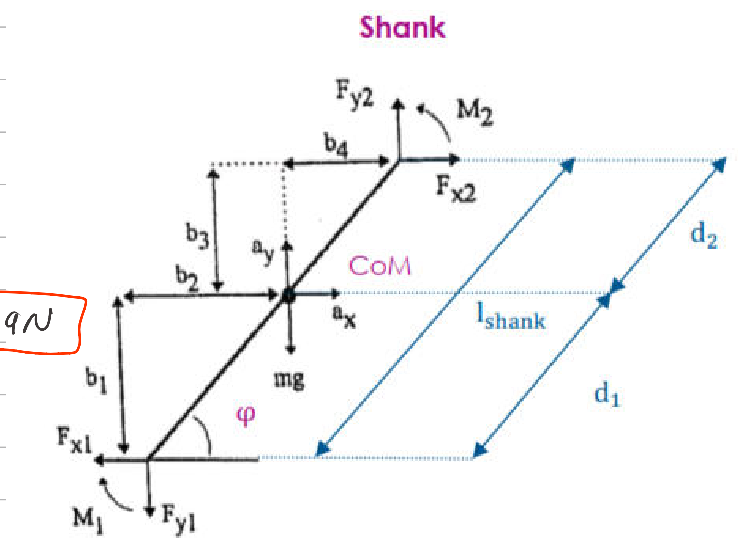
and $CoG = \dfrac{\text{Gyroradius}}{\ell} = 0.302$, $\quad r = COG \cdot \ell_{shank} = 13.59 \text{ cm}$

$$\text{Inertia} = \underset{\downarrow\, 3.72}{m_{shank}} \cdot \underset{\downarrow\, (13.59 \times 10^{-2})^2}{r^2} = 0.0687 \text{ Kg} \cdot m^2$$

$$\sum_i F_x = m_{shank} \cdot a_x$$

$$F_{x2} - F_{x1} = m_{shank} \cdot a_x$$

$$F_{x2} = 10.44 + 3.72 \cdot (-0.2) = \boxed{9.69 N}$$



$$\sum_i F_y = m_{shank} \cdot a_y$$

$$F_{y2} - F_{y1} - m_{shank} \cdot g = m_{shank} \cdot a_y$$

$$F_{y2} = 3.654 + 3.72 \cdot (9.8 - 4.2) = \boxed{24.486 \text{ N}}$$

$$\sum M = I_{shank} \cdot \alpha_y \longrightarrow M_2 - M_1 - F_{x1} \cdot b_1 + F_{y1} \cdot b_2 - F_{x2} \cdot b_3 + F_{y2} \cdot b_4 = I_{shank} \cdot \alpha$$

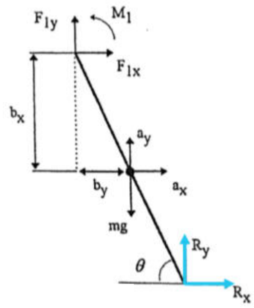
$\hookrightarrow$ $M_2 = 2.254 \text{Nm}$

$M_2 > 0$ → Knee extension thanks to quadriceps muscles

REQUEST AND SOLUTION

3.2. Compute the muscular moment and the reaction forces at the ankle when $\theta = 80°$ when:

$R_x = 160N \quad R_y = 766N \quad a_x = 3.5\frac{m}{s^2} \quad a_y = 1.8\frac{m}{s^2} \quad \alpha = -4.6\frac{rad}{s^2}$



in this case we are in the pushing phase of the foot

x components

$F_{x1} + R_x = m_{foot} \cdot a_x \longrightarrow F_{x1} = 1.16 \cdot 3.5 - 160 = -155.94N$

$F_{y1} + R_y - m_{foot} \cdot g = m_{foot} \cdot a_y \longrightarrow F_{y1} = -752.54N$

$M_1 - F_{x1} \cdot b_x - F_{y1} \cdot b_y + R_x \cdot b_x + R_y \cdot b_y = I_{foot} \cdot \alpha \longrightarrow M_1 = -57.61Nm$

CoM is the same in distances.

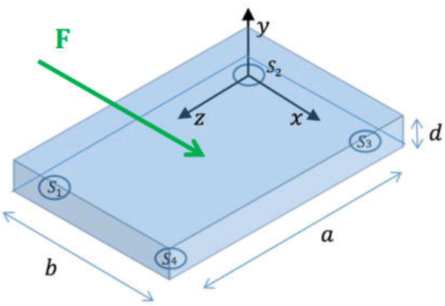
$M_1 < 0 \rightarrow$ Ankle plantaflexion thanks to triceps sural

EXERCISE 4 – Exam June 28th 2016, exercise 1

Assume to measure with a force platform the force reactions at the foot of a subject weighting 75 Kg during walk. The data collected by the platform are:

$R_{1y} = 180N \quad R_{2y} = 190N \quad R_{3y} = 220N \quad R_{4y} = 230N \quad R_{14x} = 60N \quad R_{23x} = 50N \quad R_{12z} = 10N \quad R_{34z} = 20N$

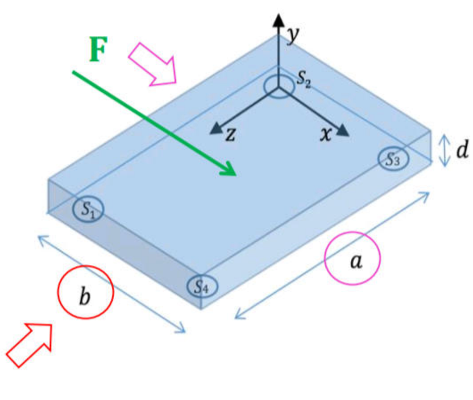
The platform is parallelepiped-shaped with a thickness d = 4 cm and rectangular base. The load cells are longitudinally spaced by a distance a = 60 cm and transversally spaced by a distance b = 40cm. All the measurements provided by the load cells refer to the lower base of the platform. The xz plane of the reference frame lies over the lower base, with the origin of the axis placed over second load cell (see figure).

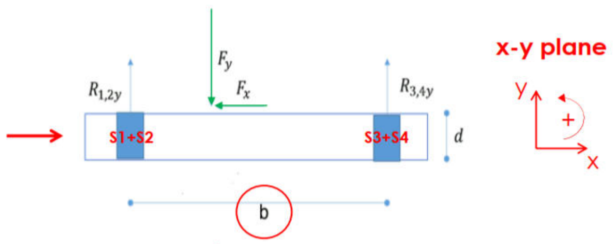


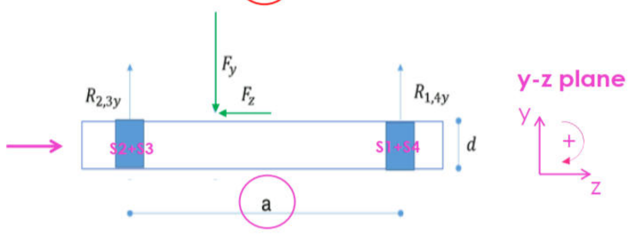
we decompose it as a 2D problem

FORCE PLATFORMS

EXERCISE 4



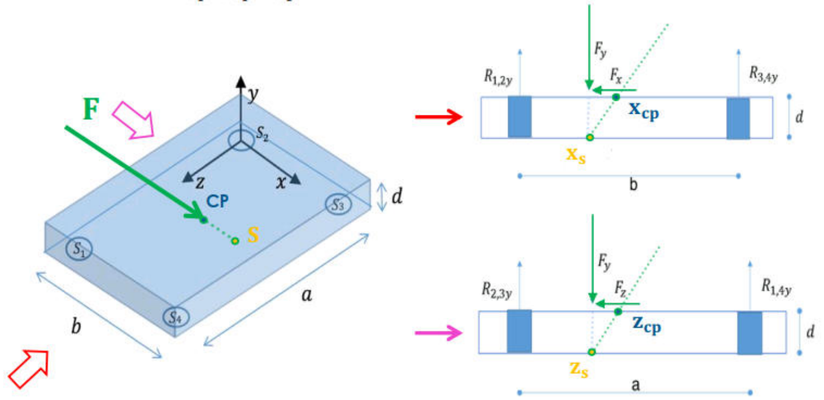




REQUESTS

4.1. Compute the location (x_s, y_s, z_s) of point S as the intersection point between the direction of the reaction force vector R and the sensor plane (lower base).

4.2. Compute the location (x_{cp}, y_{cp}, z_{cp}) of the Center of Pressure (CoP) the platform surface (upper base).



For finding the point S we need the equilibrium of the moment

SOLUTION 4.1.

$\sum M = 0$ Basically the force applied and its distance from the reference

$$-F_y \cdot x_s + R_{34y} \cdot b = 0$$

$$x_s = \frac{R_{34y} \cdot b}{F_y} = 21.95 \text{ cm}$$

$$F_y \cdot z_s - R_{14y} \cdot a = 0$$

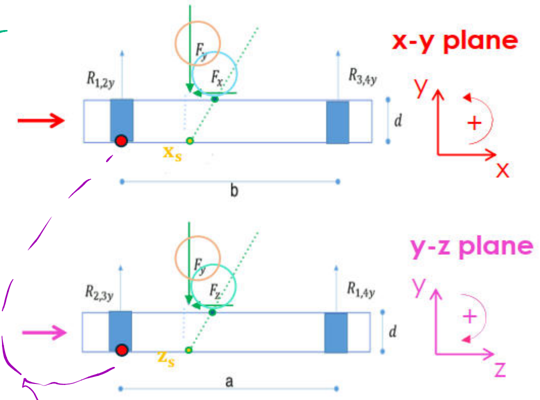
$$z_s = \frac{R_{14y} \cdot a}{F_y} = 30 \text{ cm}$$

$$F_x = R_{1x} + R_{2x} + R_{3x} + R_{4x} = 110\text{N}$$

$$F_y = R_{1y} + R_{2y} + R_{3y} + R_{4y} = 820\text{N}$$

$$F_z = R_{1z} + R_{2z} + R_{3z} + R_{4z} = 30\text{N}$$

$P_s(21.95, 0, 30)$ cm



Reference system is at the level of S2

Now we have x_s, z_s we also need to compute the ones at the CoP x_{cp}, z_{cp}

SOLUTION 4.2.

$$\boxed{\frac{d}{\Delta x} = \frac{R_{1234y}}{R_{1234x}} \longrightarrow \Delta x = d\frac{R_x}{R_y} = 0.536 \text{ cm}}$$ x_{cp}

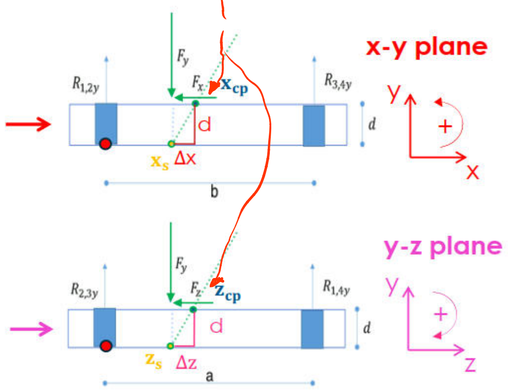
$$\boxed{\frac{d}{\Delta z} = \frac{R_{1234y}}{R_{1234z}} \longrightarrow \Delta z = d\frac{R_z}{R_y} = 0.146 \text{ cm}}$$ z_{cp}

$$x_{cp} = x_s + \Delta x = 21.95 + 0.536 = 22.48\text{cm}$$

$$y_{cp} = d = 4 \text{ cm}$$

$$z_{cp} = z_s + \Delta z = 30 + 0.146 = 30.146 \text{ cm}$$

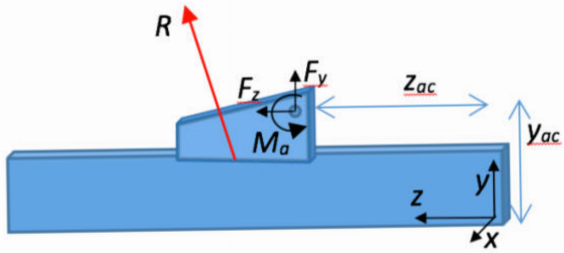
$\boxed{\text{CoP}(22.48, 4, 30.146) \text{ cm}}$



REQUESTS 4.3

The subject is standing on the platform, with a foot steady on the upper surface (no acceleration acting on it) and the other in swing phase. The axis of the ankle joint is orthogonal to the sagittal plane and passes through the point with coordinates $y_{ac} =$ 12cm and $z_{ac} = 20$ cm with respect to the platform reference frame. The center of pressure cp $[22.48, 4, 30.146]$cm lies on the sagittal plane and the mass of the foot can be considered negligible compared with the whole body mass.

Determine the reaction forces F_y, F_z at the ankle joint and the muscular moment M_a.

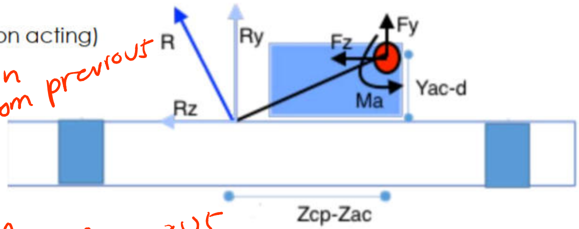


SOLUTION

Mass negligible

$$\sum F_y = m \cdot a_y \longrightarrow F_y + R_y - \cancel{mg} = 0 \quad \text{(No acceleration acting)}$$

$$\hookrightarrow F_y = -R_y = -820\text{ N}$$; obtain from previous



$$\sum F_z = m \cdot a_z \longrightarrow F_z + R_z = 0$$

$$\hookrightarrow F_z = -R_z = -30\text{ N}$$; obtain From previous

$$\sum M_a = I \cdot \alpha = 0 \longrightarrow M_a - R_z(y_{ac} - d) - R_y(z_{cp} - z_{ac}) = 0$$

$$\hookrightarrow M_a = R_z(y_{ac} - d) + R_y(z_{cp} - z_{ac}) = 30(0.12 - 0.04) + 820(0.3029 - 0.2) = 86.78\text{ Nm}$$